EXP-ECDLP-6dacdb run report

RUN-ECDLP-c770f6 · RUN-ECDLP-393d10 · 2026-10-17

This report records what ran and what the runner recorded. It states no verdict, support, weakening, significance or priority; those belong to the evidence review. A failed process is an operational fact, not a mathematical refutation.

Identity

- Goal: `GOAL-ECDLP-001` · Hypothesis: `H-ECDLP-bd1572` · Task: `TASK-20261010-ab5786` (claim epoch 1, owner `executor-ecdlp-h3-2y4itb`)
- Approval recorded in the plan: `DEC-20261010-81ff18`
- Plan: `experiments/EXP-ECDLP-6dacdb/trial-plan.json`, trials `part-a-embedding-degree-census` and `part-b-deployed-k-and-threshold`
- Run directories: `experiments/EXP-ECDLP-6dacdb/runs/RUN-ECDLP-c770f6/` (part A), `.../runs/RUN-ECDLP-393d10/` (part B)
- Code commit at launch: `c31fa9fa67cac786445d3f146bf3ed789d34049b`; sources verified by the supervisor against the plan's `source_sha256`.

Command and host

```
python3 tools/experiment_execution.py run --plan experiments/EXP-ECDLP-6dacdb/trial-plan.json --owner executor-ecdlp-h3-2y4itb --epoch 1
→ python3 experiments/EXP-ECDLP-6dacdb/implementation/run.py --part A --seeds 1 2 3 --curves 64 --out <run_dir>
→ python3 experiments/EXP-ECDLP-6dacdb/implementation/run.py --part B --out <run_dir>
```

Host: Linux cloud container (4 cores, 15 GiB), CPython 3.13.16. Declared memory ceiling 512 MiB per trial; watchdogs 3600 s (A) and 1800 s (B). Part A seeds 1, 2, 3; part B deterministic.

Trials

Trial	Run	Supervisor status	rc / check rc	Wall (s, recorded)
part-a-embedding-degree-census	RUN-ECDLP-c770f6	invalid_output	0 / 0	62.3
part-b-deployed-k-and-threshold	RUN-ECDLP-393d10	invalid_output	0 / 0	2.9

Requested 2, processes completed 2, failed 0, timed out 0. Both drivers exited 0; the independent checker exited 0 on both ("parts A, 12 curve row(s) recounted independently, 0 error(s)"; "parts B, 0 curve row(s) recounted independently, 0 error(s)" — part B's check re-verifies [l]G = O and the Hasse bounds rather than recounting curves). The supervisor recorded `invalid_output` for both with "manifest.yaml fails the run schema (tools/validate_ledger.py): expected top-level key 'run'": the driver writes a flat manifest, not the nested `run:` record the ledger schema requires; coverage reads `needs_reconciliation` (exit 3). This is a record-format defect of the frozen implementation, not a property of the measured data.

Part A, recorded values (RUN-ECDLP-c770f6)

72 cells: $d \in \{2, 3, 4, 5\}$, two arms (special 16-bit $p = m^d - c$; seed-matched random 16-bit prime), 3 primes per d per arm, 3 seeds; 64 random curves per cell with the largest prime factor l of the point count and the embedding degree $k = \mathrm{ord}_l(p)$ computed exactly; anomalous curves (point count equal to p) excluded and counted (4 cells record `anomalous: 1`); a per-cell `ctrl_count` row records a Hasse check and `point_killed: true`.

- Curves: special 2300, random 2304 (recorded `n_curves`).
- `mean_log2_t` with $t = (p - 1)/k$: special 1.2487, random 1.2667, difference −0.0181, permutation p-value 0.684.
- `small_k_fraction`: special 0.1591, random 0.1328, difference 0.0263, recorded standard error 0.0104.
- t histogram, bins [1], [2], [3–4], [5–16], [≥17]: special (851, 725, 288, 309, 127), random (862, 679, 304, 330, 129); chi-square 2.71, permutation p-value 0.612.
- `smallest_k`: both arms record $k = 1$ at $l \in \{5, 7\}$ and at $l = 11$ (random) or $l = 29$ (special).
- Supersingular controls: 43 rows, every one records $k = 2$ (l from 19 to 15671).

Part B, recorded values (RUN-ECDLP-393d10)

Six public parameter sets; prime checks on p and l, Hasse check, an order check [l]G = O where a base point was available, an exact embedding-degree lower bound by direct exponentiation to 10^6, the rho cost $\log_2$ of the square root of $\pi l/2$ and k^* under three recalled $L(1/3, c)$ constants with the $o(1)$ dropped (c = 1.923 GNFS, 1.747 exTNFS, 1.526 SNFS/STNFS).

curve	special	l bits	k measured	rho log2	k* GNFS	k* exTNFS	k* SNFS	[l]G = O
Curve25519	yes	253	> 10^6	125.8	9	11	16	true (ladder)
P-256	yes	256	> 10^6	127.8	9	12	17	true
secp256k1	yes	256	> 10^6	127.8	9	12	17	true
P-521	yes	521	> 10^6	260.3	27	34	48	true
Ed448	yes	446	> 10^6	222.8	21	27	37	not run (no base point recorded)
brainpoolP256r1	no	256	> 10^6	127.5	9	12	17	true

Embedding-degree searches took 0.28–0.44 s per curve (recorded). The runner's own note: constants are recalled and the o(1) is dropped, so k* is optimistic for the attacker; k measured is an exact lower bound.

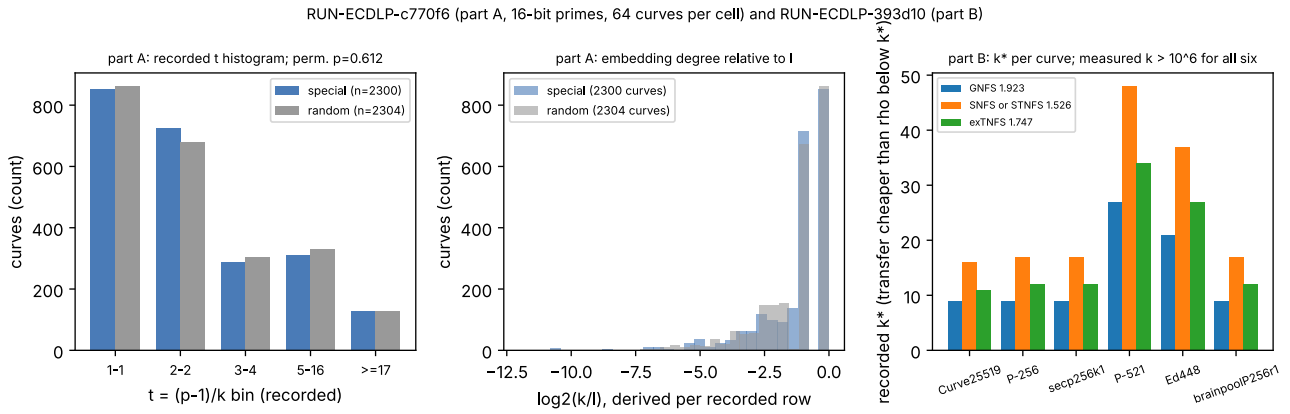


Figure 1: Left: recorded part A t histogram per arm (RUN-ECDLP-c770f6, 2300 special and 2304 random curves). Middle: $\log_2$ (k/l) per recorded curve row, derived as the quotient of the recorded k and l. Right: recorded part B k* per curve and constant (RUN-ECDLP-393d10). Data: `data.json`, drawn by `plot.py`.

Missing data

Ed448 has no recorded base point, so its order check is `null`. No per-curve CPU time was recorded.

Notes

Both run directories hold the supervisor's ten artifacts plus, for part B, `deployed-table.md`. `data.json` copies the recorded test block, the supersingular controls, the per-cell (k, l, t) rows and the whole part B block from the two `raw-result.json` files.